

6(a)	ligase joins / AW, Okazaki fragments / fragments on lagging strand / AW ; catalyses, formation of phosphodiester bonds ; <i>to join, Okazaki / DNA fragments</i> <i>plus max 2 from:</i> lowers / AW, activation energy ; provides alternative reaction pathway ; (mode of action) lock and key / induced fit / formation of enzyme-substrate complex ; <i>allow described in terms of joining adjacent nucleotides</i>	3
6(b)	G₁ and S and G₂ ; A mentioned anywhere in the answer and in any order A synthesis for S <i>plus any three from:</i> cell growth in G₁ and G₂ ; A cell / cytoplasm volume, increases in size (high levels of) protein synthesis / AW, in G₁ (and some protein synthesis in G₂); organelles replicate in, G₁ / G₂ / G₁ and G₂ ; A examples e.g. mitochondria DNA replication in S phase ; A formation of, sister, identical, chromatids centrioles, replicate / complete replication, in G₂ ; A procentriole formation / centrioles begin to form, in S phase AVP ; e.g. histone synthesis in G₁ / S, phase G₂ phase increase in energy stores G₁ checkpoint, qualified e.g. to proceed to S phase G₂M, checkpoints occur / checkpoint before mitosis begins bod DNA is checked before mitosis	4
6(c)(i)	metaphase ;	1

Question	Answer	Marks
6(c)(ii)	cell plate formation cell wall formation ; A spindle / spindle fibre, disassembly A formation of cell surface membrane <i>in context of forming daughter cells</i> bod cytokinesis	1